The second experiment introduces Treatment First in the full calibrated model, with the externalities on children, in three steps. Each step lets one more object respond, so that each isolates one thing. The first two repeat the first experiment's two runs, so that each has a counterpart without the externalities. First, the drug price and πV are both held at their Housing First levels. Only the screen then acts, as in the first experiment's first run, and the one difference between the two runs is whether drug use harms children, so comparing them shows what the externalities do to the verdict. Section 6.3 then isolates the externality itself within each run, as the part of the newborn's gain its parent does not count. Second, πV clears the budget, as in the first experiment's second run, so that the screen's savings are spent on vouchers for others. This shows what that spending is worth once the children are counted, with its externality separated the same way. Third, the drug price clears the drug market as well. The drug price comes last because its response is a pure general-equilibrium price effect: the screen lowers the demand for the drug, and the cheaper drug draws users back. With both objects responding, as they would after a real reform, this step is also the main experiment: whether Treatment First raises welfare. Table 5 also shows, for both experiments, the drug price clearing with πV held.

The third experiment is the transition from Housing First to Treatment First in the full model, with both the drug price and πV clearing the drug market and the budget, respectively, every period (Section 6.4). The steady states are decades apart, and the households that have a SUD when the reform lands can be sure of keeping their assistance only by entering treatment and stopping use, while the disorder itself clears only as fast as recovery allows. The transition establishes how large the early costs of the reform are, who bears them, and whether children born at the reform gain.

After the main transition, I explore alternative transitions in which households with a SUD cannot stop using (Section 6.5): the screen with no exemption for treatment from the first day of the reform, the same screen announced five periods before it starts, and the same screen with treatment alone keeping the voucher for its first five periods, each at two detection rates. Section 6.1 defines the welfare criterion all three use.

The screen. The detection rate pdet is the chance, each year, that a voucher holder who uses drugs is caught. It sets how much using costs a holder, and so how strongly Treatment First deters use: as it rises, more users stop, and those who keep using lose their voucher sooner. What the rate captures is how often a household is tested, which no programme reports. I therefore set it so that the screen deters use as much as a drug test with a severe penalty does in the closest evidence, the US military's random testing (Section 2). Among service members who stayed after the military began testing at random and dismissing those who failed, past-year marijuana use fell by about 61 percent of what it would otherwise have been (Mehay and Pacula, 1999).34

The model's counterpart applies Treatment First to the benchmark's households, holding the distribution, the drug price and πV at their Housing First levels, and follows each user subject to the screen, that is, each user who holds a voucher or is eligible and offered one. With pdet $=$ 20 percent, 62 percent of them stop when

34That is the raw comparison among those who stayed, which leaves out the users the testing dismissed. Mehay and Pacula's own estimates of the policy's effect, which allow for who left, are 4 to 16 percentage points of the 35.7 percent who used in 1980, a fall of 11

to 45 percent. At half the calibrated rate, 41 percent of the users subject to the screen stop, inside that range (footnote 36).

the screen is introduced,35 and none starts.36 A household that does not use, and is in treatment if it has the disorder, is never flagged, since a positive screen is confirmed in a laboratory before it has consequences.

The release rate ψx is set to 1/3, so that the expected exclusion is three years. That is the same period for which the Housing Choice Voucher programme already bars a household evicted from federally assisted housing for drug-related criminal activity (Code of Federal Regulations, Title 24, Section 982.553). The test cost ctest is set to 0.031 percent of average earnings in the economy, the equivalent of $35 a test. The Texas Legislative Budget Board's fiscal note for Senate Bill 11 puts a test at $35 (Legislative Budget Board, 2013), the upper end of the estimates, and taking the upper end again loads the experiment against Treatment First.

6.1 The welfare criterion

Every welfare number in this paper is a consumption-equivalent variation (CEV). For a newborn it is com- puted as follows. Write V b for expected lifetime value under the benchmark and V c for expected lifetime value under a counterfactual policy. Both are taken at birth, before any of the newborn's draws are realised, including the household it is born into. Each is taken under its policy's own distribution of newborns, which the policy moves, since a child's starting productivity depends on what happened to its parent.37 The CEV Δis the proportional change in non-housing and non-drug consumption, applied at every date and in every state, that makes a newborn under the benchmark as well off as the same newborn under the counterfactual:

V b(

(1 + Δ) c

) = V c(c). (30)

A positive Δtherefore means the counterfactual is worth that much of a permanent consumption gain, and a negative one that it costs that much.

Whose welfare. Welfare here is measured for newborns, and it includes childhood in the parent's house- hold and adult life after it. For a child born in period t it is

Wt =

JK ∑

j=1

β j−1 (ξ(j) −1)

∏

S U

 Xc(j, s)

 dΦt + j − 1(j, s) Φt + j − 1(j, S) Ct

+ βJK

S Vt + JK(1, s) dΦt + JK(1, s)

Φt + JK(1, S) At

, (31)

where Vt(j, s) and Φt(j, s) are the value and the distribution of Section 4.10 and Section 4.13 in period t, the subscript being the date and j the age, and Xc(j, s) is the bundle of (11) that a child shares with a parent of age j in state s. The state in both integrals is the parent's: a child at home has no state of its own, its bundle is its parent's, and the state it enters adult life with is drawn at JK by N.

The first term, Ct, is childhood. A household joins the workforce at $j = 1$ with its child just born, so the households of age j in period $t + j - 1$ are the surviving parents of the children born in period t, each child in its j-th year at home. The child's flow utility from the bundle it shares with its parent is averaged over the parents alive that year, weighted by $\xi(j) - 1$, the child's weight in the household's own equivalence scale, and discounted to the child's birth. Its parent counts the same flow at $\omega c(\xi(j) - 1)$ in (24), so only ωc separates the parent's valuation from the criterion's: the parent internalises the share ωc of its child's childhood and leaves the share $1 - \omega c$ external, as Section 4.10 defines it.38

35The military figure is for any past-year use of marijuana, a broader measure than the model's, whose users have a disorder of any drug, marijuana among them. Therefore, the two are compared as proportional responses.

36At half the rate (pdet = 10 percent), 41 percent of the users subject to the screen stop on impact, about two thirds of the military's response. The results hold: in the long run, with the drug price and πV clearing, Treatment First cuts drug use by 14 and homelessness by 6 percent, against 21 and 11 percent, respectively, at the calibrated rate, and a newborn is better off by 1.65 percent of lifetime consumption, against 2.57.

37The ex-ante expected newborn welfare is the standard criterion in this literature (Conesa et al., 2009; Guvenen et al., 2023). 38A child whose parent dies before JK stays in the average: for the rest of its childhood it is valued as the average child of its cohort rather than sent into adult life at the parent's death, as the law of motion (29) sends it, because an orphan has no consumption or dwelling of its own in the model. Sending it into adult life at the parent's death instead, with the probability χ(j) of (24), scores Treatment First lower: adult life carries the flow value of being alive אand childhood does not, so that convention credits a child for losing its parent early, and Treatment First keeps more parents alive.

The last term, At, is adult life, which begins at JK: it is the average value at $j = 1$ of the households that enter in period t + JK, placed by N as in (29), discounted by the JK years

spent at home. The parent's own value, which Section 6.3 sets beside it, carries no such discount: the parent is at $j = 1$ in the year its child is born, so the parent's value and the child's childhood are dated alike. The value at $j = 1$ in At is the one the child's own problem maximises, so the child's own child enters there at the weight ωc.

Counting the child's childhood at its own weight, $\xi(j) - 1$, rather than at its parent's $\omega c(\xi(j) - 1)$ is what brings the externality on the child into welfare; Section 6.3 separates that externality from the rest of the gain. In a steady state the dates drop out.

Two horizons are used, and they are the same object measured at different dates: a newborn on the date the policy is implemented, who lives through the transition, and a newborn in the new steady state.

6.2 The long run

Table 5 sets the long run under Treatment First beside the benchmark, and, in the first five columns, Housing First and Treatment First without the externalities.

Treatment First without externalities. To capture the no-externality case, I treat the child as appear- ing only as an adult: its parent does not weigh it, its productivity does not depend on what its parent did, and its childhood, the years its parent is 26 to 50, is not counted in welfare. Every other parameter keeps its calibrated value: the economy is not refitted, so that the comparison changes nothing but these three switches.

With the drug price and πV held at their levels under Housing First, Treatment First cuts drug use by 28 percent and the share with a SUD by 27 percent, against Housing First with the externalities switched off, while homelessness falls by 5 percent (first two columns of Table 5). And yet, no household is better off under Treatment First, with newborns losing 1.17 percent of lifetime consumption as adults. These losses capture the value of the option removed from a rational, forward-looking agent.